



Society of Petroleum Engineers

SPE-230221-MS

Revitalizing Aged Wells with Complex Issues Using Cost-Effective, Rigless Solutions and Low Co2 Emissions

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Copyright 2025, Society of Petroleum Engineers DOI [10.2118/230221-MS](https://doi.org/10.2118/230221-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

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Abstract

The field presented here is a giant heterogeneous carbonate field, located in an offshore Abu Dhabi. It has multi-stacked highly heterogeneous sub-reservoirs formed by regressive cycles of sedimentation & separated by partial or complete barriers, consists of a very large rich gas cap with oil rim. The current development plan is oil production from the oil rim with peripheral water injection & crestal gas injection. The reservoir dip angle is 3 degrees, and it has been producing since 1962. Permeability in the reservoir layers ranges from about 1 mD to a thousand mD and porosity ranges from 10 to 25%. Sealine export pressure is quite high (almost 1100 psia at the well-head) and there is no artificial lift. Around 6% of the strings are inactive in the field. Some strings are inactive due to high GOR and others are unable to flow (UTF) due to high water production. Some other strings are shut-in for integrity or flow assurance reasons. Overall GOR guideline is currently not to exceed 3500scf/bbl., water cut limit that makes wells UTF is quite low due to the high sealine pressures mentioned previously.

Aged wells with complex integrity concerns present significant operational challenges, factors such as old well design, material selection, and offshore environmental conditions contribute to these issues. Addressing these wells can increase operational costs and lead to production interruptions and increased greenhouse gas emissions. An engineering team, including members from reservoir, petroleum, production, and well integrity divisions, conducted an assessment of these wells to identify efficient solutions for integrity issues. Two wells were selected for further analysis and proposal development. These wells function as dual oil producers and exhibit challenges related to high gas-oil ratio (GOR), downhole safety valve (DHSV) malfunction, and sustain annulus pressure (SAP).

Sustained casing pressure (SCP) is one of the primary well integrity concerns. Multiple annular pressures are present in the wells. Well integrity teams aim to maintain all casing pressures below the Maximum Allowable Annulus Surface Pressure (MAASP) during steady state operation. If annular pressure exceeds this threshold, the casing displays unacceptable SCP levels. Unmanaged SCP may result in adverse outcomes if not addressed promptly.

This paper outlines a rigless approach to managing complex well integrity issues, aiming to maintain production while minimizing environmental impact and carbon emissions.

Introduction

During the production life cycle, completion component failures such as tubing hanger, DHSV, tubing, or packer leaks can cause unwanted pressure (SCP) or barrier loss, posing risks to well production, safety, assets, and the environment. These issues may damage primary well-integrity barriers and result in loss of containment. If the invading fluids contain corrosive elements like H_2S or CO_2 , they can further degrade integrity barriers, making previous MAASP values unreliable.

One of the principal challenges associated with ageing wells is that their barriers are often situated thousands of feet beneath the earth's surface, making integrity assessments particularly complex. Cement is utilized as the primary physical barrier, sealing off fluid migration from the wellbore to unintended zones and mitigating the risk of surface leakage. The effectiveness of this seal is highly dependent on in-situ environmental conditions, cement composition, and wellbore stresses. Casing corrosion remains a prevalent issue due to the inherently corrosive nature of reservoir fluids and is frequently unavoidable. Furthermore, certain surface equipment has become obsolete over time, presenting difficulties in sourcing appropriate replacement parts.

Continuous monitoring and evaluation of all wells should be conducted using the following data sets:

1. Well construction and operational data, including:
 - Formation characteristics (lithology, cap rock properties, pore/fracture pressure gradients, etc.).
 - Reservoir fluid properties.
 - As-built well attributes such as casing, tubing, and completion components.
2. Well construction documentation, such as drilling and completion fluids records, cementing reports, and logs.
3. Operating limits established during well handover.
4. Operational parameters, including production/injection rates, pressures, and temperatures.
5. Corrosion and erosion assessments.
6. Diagnostic test results and maintenance records.
7. Historical annuli pressure measurements.

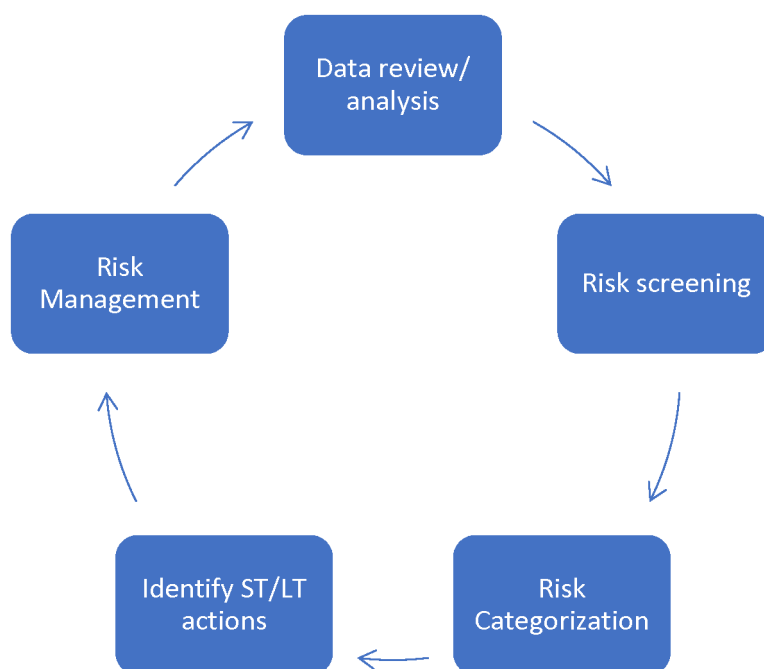


Figure 1—Well Integrity Management cycle

The existing methods of mitigating annulus pressure and completion failure can be technically complex, very costly (in terms of time, equipment, environmental impact & loss of production).

According to [ISO standard 16530-1:2017](#), Sustained Casing Pressure (SCP) is defined as:

- Unintended pressure in a contained annulus resulting from pressurized formation fluids (liquid or gas).
- Measurable at the wellhead termination of casing annulus valves.
- Pressure that rebuilds after being bled down.
- Not caused by wellbore temperature fluctuations (thermal effects).

Wells affected by SCP are often shut-in, leading to production deferrals and, in some cases, eventual plugging and abandonment. SCP typically results from seepage of subsurface fluids.

As per ADNOC well integrity standards, all wells that are capable to flow must be equipped with standards down hole safety valves, any other type of DHSV not surface controlled are not accepted as they are not reliable in case of emergency cases.

The objective of this project was to address these challenges cost-effectively while safely returning wells to production. Conventional remediating methods of such wells can result in substantial costs for operators, disrupt production operations, and generate CO₂ emissions.

To address these issues, an engineering team composed of experts from reservoir, petroleum, production, and well integrity divisions collaboratively assessed affected well. Their goal was to study and propose an innovative approach to efficiently resolve well integrity concerns. Two wells were selected for detailed analysis due to combined challenges of high gas-oil ratio (GOR), downhole safety valve (DHSV) failure, and sustained annulus pressure (SAP).

The team established rigorous procedures and guidelines for identifying suitable candidate wells. After reviewing the operational history, they developed specific solutions, focusing on isolating problematic strings without rig intervention and diverting flow to healthy strings. They also repaired failed DHSV control lines using sealant injection. All interventions were performed from a wellhead tower, eliminating the need

for additional equipment such as rigs or barges, thus improving both efficiency and effectiveness during the pilot project.

Following comprehensive monitoring and integrity assurance tests, both wells were successfully brought back online. This resulted in an increase of over 5 MBD in oil production, a reduction of more than 1,400 tons of CO₂ equivalent greenhouse gas emissions, and optimal utilization of drilling units for other priority wells, all achieved with full compliance to HSE standards.

This multidisciplinary approach not only focused on reservoir and integrity management but also demonstrated significant gains through creative problem-solving and teamwork. The success of both operations highlights' exceptional collaboration across teams, delivering high-quality outcomes while mitigating well integrity risks. Conducting the trial offshore required meticulous planning due to increased complexity and demonstrated the viability of this novel intervention method for similar future applications.

By improving integrity, the solution reduces the potential for loss of well control and minimizes the risk of premature plugging and abandonment. It provides an effective mitigation strategy that enables continued production until scheduled workovers, while eliminating CO₂ emissions associated with rig-based interventions. In comparison, conventional SCP remediation methods are costly, disruptive, risky, and do not guarantee successful outcomes.

Treatment Overview

The first well which was suffering from SAP -A and nonstandard DHSV was cured by securing the communicating string and then divert the flow to the healthy string following below operation steps.

First well project

Well X was drilled in 2001 and completed in 2011 as dual oil producer; the well was CAT-2 due to the substandard DHSV installed A3 valve in the short string

The short string was producing ~3000 BPD with high GOR while the long string was unable to flow with high GOR, also the long string had obstruction at 7832 ft complicating the intervention

Due to the above integrity concerns, the well was kept shut in, placed in drilling bar chart for rig workover

The first well project was conducted by plugging the short string that has substandard DHSV (A3) producing high GOR

Then divert the flow to long string restoring the production of 3MBD:

The job outline steps were as follows:

- Plug Short string with mechanical plug utilizing the slick line unit on the tower (no barge or rig), isolate Y shallow reservoir
- Set bridge plug in the long string at 7802, isolate (W) deep reservoir
- Perform 5 shot of mechanical punch in the long string against the shallow (S) reservoir to divert the flow from short string to healthy long string with standard functioning DHSV
- Installed new DHSV in the long string.
- Flow the well and perform the flow test

It is necessary to implement certain modifications ([Fig2](#)). to the proposed operational procedures with the reservoir and petroleum teams and integrity recommendations

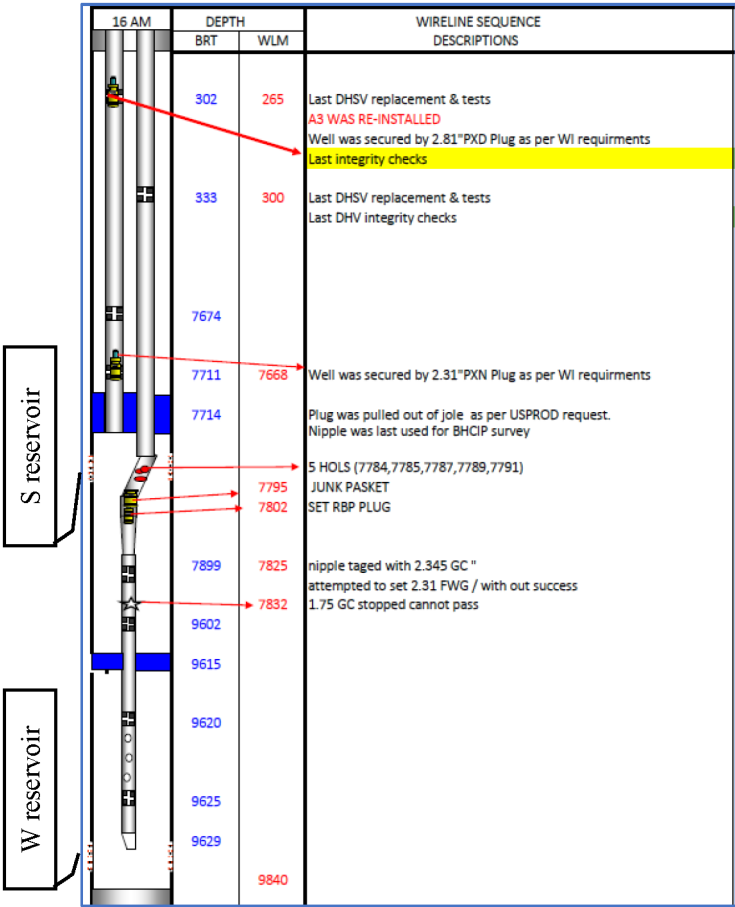


Figure 2—well X completion and operation steps

Formation Name	Depth (Ft)		13 3/8"	9 5/8"	7" Liner	4 1/2"
	Top	Bottom	NA	SBT-VDL	SBT-VDL	SBT-VDL
Shallow formations (A)	1126	1192	Not Logged			
	1192	1921	Not Logged			
	1921	2200	Not Logged			
	2200	3282	Not Logged			
	3282	4000	Not Logged			
	4000	4060	Not Logged			
	4060	4280		Possible Communication		
	4280	4509		Possible Isolation		
B	4509	4839		Possible Isolation		
C	4839	4909		Possible Isolation		
D	4909	5402		Possible Communication		
E	5402	5720		Possible Communication		
	5720	5885		Possible Isolation		
F	5885	6275		Possible Communication		
G	6275	6300		Possible Isolation		
	6300	6450		Possible Communication		
	6450	6499		Possible Isolation		
H	6499	6612		Possible Communication		
I	6612	7279		Possible Isolation		
J	7279	7549		Possible Isolation		
K	7549	7700		Possible Communication		
	7700	7720		Possible Isolation		
	7720	7820		Not logged		
	7820	8001		Not logged	Possible Communication	
	8001	8050			Possible Communication	
	8050	8088			Possible Isolation	Possible Isolation
L	8088	8715			Possible Isolation	Possible Isolation
M	8715	8953			Possible Isolation	Possible Isolation
N	8953	8986			Possible Isolation	Possible Isolation
O	8986	9030			Possible Isolation	Possible Isolation
	9030	9071			Possible Isolation	Possible Isolation
P	9071	9100			Possible Isolation	Possible Isolation
	9100	9145			Possible Isolation	Possible Isolation
Q	9145	9160			Possible Isolation	Possible Isolation
	9160	9194			Possible Isolation	Possible Isolation
	9194	9292			Possible Isolation	Possible Isolation
R	9292	9403			Possible Isolation	Possible Isolation
S	9403	9460			Possible Isolation	Possible Isolation
	9460	9475			Possible Isolation	Possible Communication
T	9475	9521			Possible Isolation	Possible Communication
U	9521	9605			Possible Isolation	Possible Communication
V	9605	9671			Possible Isolation	Possible Communication
W	9671	9745			Possible Isolation	Possible Communication
	9745	9765			Possible Isolation	Possible Isolation
X	9765	9800			Possible Isolation	Possible Isolation
	9800	9860			Possible Isolation	Possible Communication
	9860	9878			Possible Communication	Possible Communication
Arab D5	9878	9910			Possible Communication	Possible Communication
	9910	9930			Possible Communication	Possible Isolation
	9930	9964			Possible Communication	Not logged
Diyab	9964	10010			Possible Communication	
	10010	10060			Not logged	

Figure 3—Cement evaluation sheet for the well (X)

Rev:21/Dec/2012
30" conductor
@327'

13 3/8" casing
@4057'
68# J-55 Butt

Pkr@7717'
Pkr@7737'
TOL@7752'
Pkr@8005'
4 1/2" TOL @8013'
9 5/8" @8050'

7", 29# L-80

4 1/2" 13.5# L-80

PBTD 9950'
4 1/2" @10000'
7" @10057'

Second well project

Well History

- Aug. 1984: well XX was drilled and completed as a deviated dual producer from two different zones (dual strings)
- May-1993: Well was worked over and completed as dual oil producer with more perforations of additional zone (dual completion)
- Jan-2023: Short string was closed due to high annulus pressure caused by communication between Control line and production annulus and secured with Plug in the DHSV Nipple @323 FT
- Feb-2023: Long string control line was found clogged
- Mar-2023: Long string control line was cured with piston activated special type Sealant
- May-2023: Long string control line passing & DHSV (closed), well inactive due to Flapper closure

Objectives

- To Reactivate the inactive well by diverting Flow of Upper Reservoir from short string to long string after isolating the lower Reservoir followed by curing the down hole safety valve control line leak by sealant injection.

Status Pre-Job (reservoir concerns)

- Long string inactive due to reservoir issue High Water cut, represented by Q reservoir on the [Figure 5](#)
- Short string suspended due to integrity issue Communication between DHSV Control Line and A-Annulus
- assess Well integrity status by performing communication test with A-annulus and confirming the integrity of existing barriers

Well integrity concerns

DHSV control line leak was confirmed to be cause of communication to the production annulus as well as the communication from the short string through tubing joints.

Control line leaks in Downhole Safety Valves (DHSV) can significantly impact integrity and cause production disturbance. Here's how to address and repair these leaks:

1. Causes of Control Line Leaks:

- Bad connections or damaged seals.
- Parted or plugged control lines.
- Thermal effects during production cause loose connections.

2. Repair Procedures:

- Inspect all fittings and connectors thoroughly.
- Tighten connections using a standard torque wrench.
- Pressure test the control line to ensure no future leaks.
- Check for apparent damage, kinks, or pin-holes before connecting.

- Use sealant techniques to cure leaks, which polymerize at the leak site.
- For larger leaks, a workover may be required to replace the control line.

3. Preventive Measures:

- Clamping the control line to the tubing for mechanical protection.
- Encapsulating the control line to prevent damage.

Proper sealant type should be selected for sealing leaks in control lines. It is injected into the control line where it polymerizes into a flexible solid, effectively sealing the leak. This method is beneficial for small leaks but may not be suitable for larger leaks, such as those caused by parted tubing. The sealant could be of two type , first one will solidify at the leak site due to pressure differential, the second type can be as piston activated sealant which is used in this example well with successful result. This sealant is injected at a specific rate according to the leak rate of the control line and will remain in the liquid phase acting as piston on the control line leak point.

Actions taken on long string

1. Recorded BHCIP and SGS static survey
2. Flow test and confirmed water cut
3. Isolated Long string with down hole plug
4. Punched tubing above SSD (Side Sleeve Door) and successfully reactivated upper reservoir through long string
5. Flow test upper reservoir, excellent test was observed with the oil rate increase ([Table 1](#))

Table 1—Well XX flow test report

Choke (/64")	WHFP (psi)	Oil (bopd)	BS&W (%)	GOR (SCF/STB)	Remarks
24	1650	1187	2	2523	Reactivated Post Perforation
40	1510	2773	2	2699	

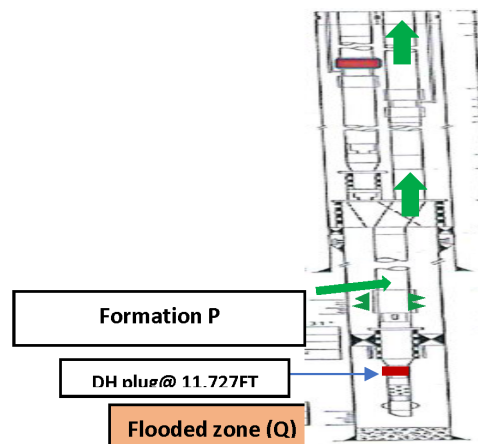


Figure 4—Well XX completion (producing from P & Q reservoir)

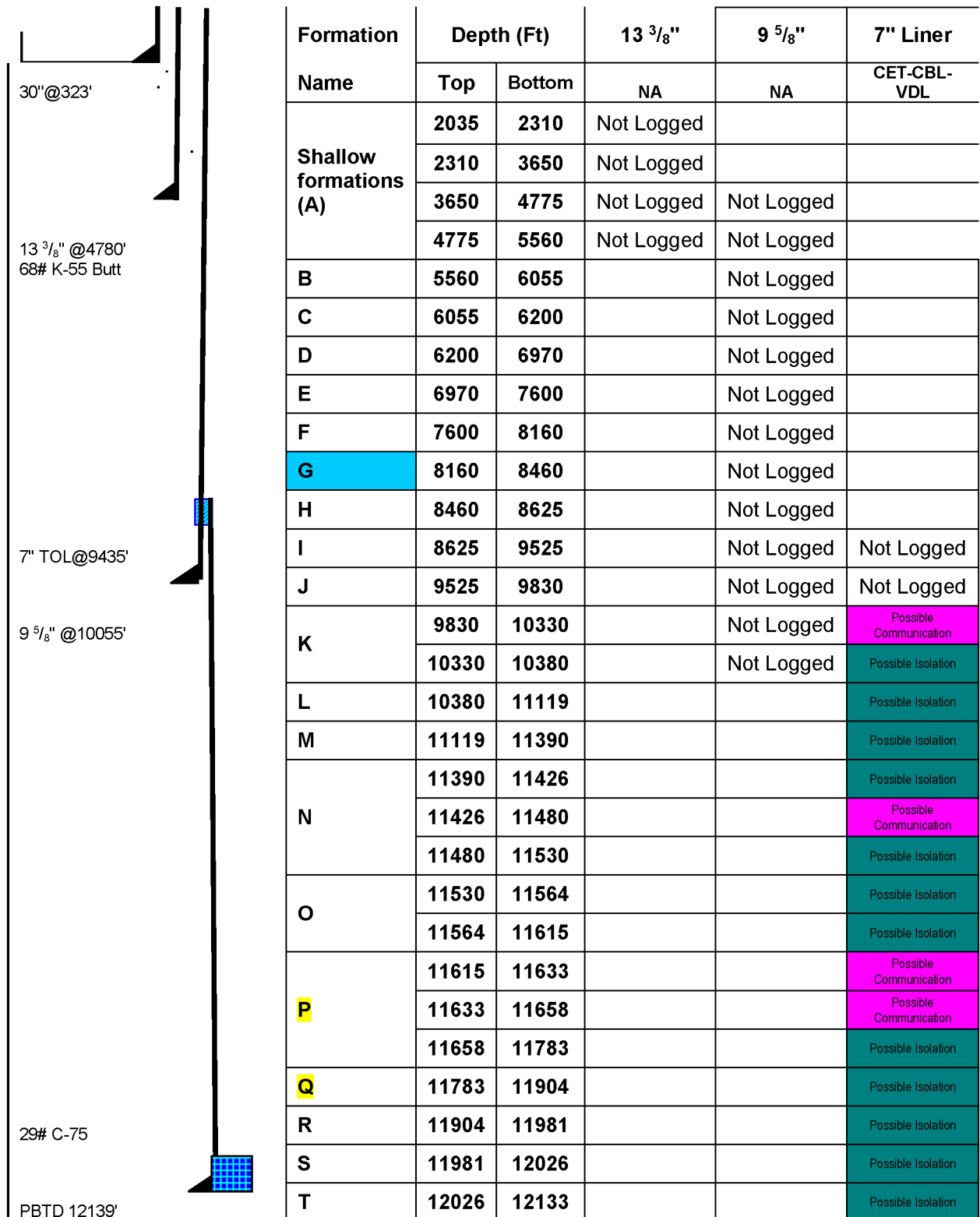


Figure 5—(Well XX cement evaluation sheet)

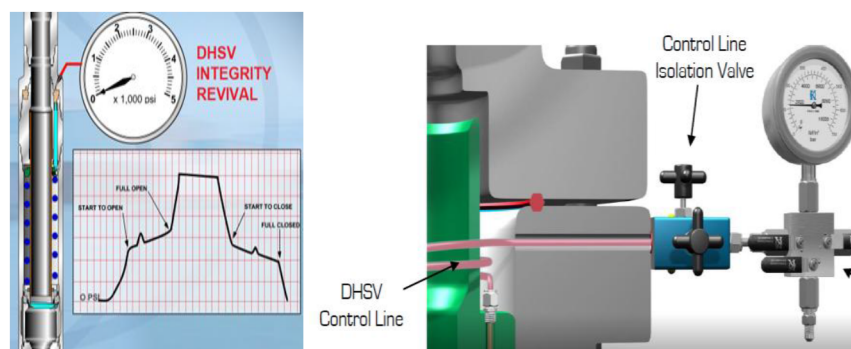


Figure 6—Reviving DHSV by piston activated sealant

Actions on the short string

1. Retrieved down hole plug from DHSV-Nipple
2. Recorded SGS survey
3. Set DH plug in XN-Nipple
4. Performed communication test for A-Annulus and inflow test DH plug (hold)
5. Retrieved DH plug and performed TBG preservation with diesel

Curing DHSV on well XX

Following the new approach of sealant injection and Ingress valve to cure the control line leak, both strings DHSV's were successfully restored and new standard DHSV was installed and tested ,below [figure.5](#)

Conclusion

The successful work to manage the reservoir and isolate the which communicates to the production annulus due to the high Gas Oil Ratio(GOR) and undesired flow(high water cut) thru the leak path of the failed completion ,riglessly perforate another interval of good reservoir characteristics to divert the flow from the non-integral string to the healthy string, followed by the rectification of the failed down hole safety valve by injecting the piston -activated sealant to replace the non-standard DHSV has proven the importance of implementing new methods to restore the well production as well as the integrity, The trial treatment project meets its objective of mitigating pressure in both wells and restoring the full functionality of the downhole safety valve through the injection of a piston-activated sealant and installation of an ingress valve to prevent sealant loss. Moreover, the gain was remarkably noticed on the rig cost saving of \$ 5-7 M per well and on the other hand resulted in reducing the GHG emission by around 670 Ton CO₂ gas reduction per well(total 1400 Ton of Equivalent CO₂ emissions)

These treatments provided a rapid and effective solution to address SCP well integrity concerns, resulting in reduced downtime and increased production availability.

Following the successful completion of work on both wells:

- They were removed from drilling rig operation plans and barge plans, yielding substantial savings in OPEX and CAPEX workover rig costs.
- The total production gain from recommissioning amounts to approximately \$226 million.
- The job can be efficiently performed on the tower at minimal cost, utilizing in-house resources (with an estimated cost of \$70,000/per job).
- This approach is suitable for application to other wells with similar conditions.
- No additional resources required rather than available field resources

Recommendation

- Assigned focal point teams must conduct a thorough job analysis prior to finalizing the way forward
- Review detailed history reports including reservoir data, well integrity data, production and completion data which is one of the key elements to support and make solid decisions.
- The final program to consider not only the best solution to restore integrity but also to have a minimum impact on the environment and ensure full compliance to the HSE rules while assuring available resources with minimum CO₂ emissions
- Field resources optimization can add great savings, especially for towers having more than a well with same problem
- Complete appropriate job risk analysis and proper material & tools selection
- Perform all required integrity tests on the down hole barriers and surface barriers
- Select the best tools & materials in the market and worldwide that can fit for purpose and solve the complex issues
- Assure the success record of the selected materials with the certifications
- Perform mockup prior to starting any field operation
- Perform multi flow tests following the job completion to ensure production enhancement
- If the treated barriers didn't pass the required integrity verifications, the job can be repeated with design modification as per the engineering team

Project Added Value

- No production downtime: Traditional rig workover activities to address SCP can result in two to three months of lost production, whereas the new technique eliminates the need to shut in the well.
- Stable production rates: Historically, shutting in a well has been associated with reduced post-intervention production rates. This alternative solution mitigates that risk.
- Enhanced well integrity: Untreated SCP may ultimately compromise well control and result in the release of hazardous substances into the environment. Such scenarios could cause total loss of the well as a productive asset and necessitate an expensive abandonment process.
- Rig availability: Rig-less intervention ensures rigs remain accessible for alternative operations, thereby optimizing resource allocation.
- Health, Safety, and Environmental (HSE) benefits: The approach reduces personnel exposure and minimizes the risk of uncontrolled subsurface fluid releases. Furthermore, the compound used is non-toxic to the environment and is suitable for both offshore and onshore applications, helping to reduce CO₂ emissions.
- Operational practicality: Implementation is feasible without causing any operational disruptions.
- Production gain of 5MBD for both wells

Acknowledgments

The authors would like to extend their appreciation and thanks to:

- ADNOC (Dr Elyes Draoui (VPFD US/NS/LZ) and Dr Abdulla Naqbi) for their support to implement the pilot jobs.
- ADNOC (Mr. Mohamed Saleh Ali /well Integrity Manager)

The authors also would like to thank:

All involved teams from well integrity, well performance management, field production operations, slick line operations team

Nomenclature

ADNOC	: Abu Dhabi National Oil Company
MAASP	: Maximum Allowable Annulus Surface Pressure
MAWOP	: Maximum Allowable Working Operating Pressure
SCP	: Surface Casing Pressure
DHSV	: Down Hole Safety Valve
SCSSV	: Surface Controlled Sub Surface Safety Valve
SGS	: Static Gradient Survey
OPEX	: Operating Expenditure
CAPEX	: Capital Expenditure
WHFP	: Well Head Flowing Pressure
BHCIP	: Bottom Hole Close In Pressure
BS&W	: Basic Sediment and Water
WC	: Water Cut
LS	: Long String
SS	: Short String
A3	: Non-Standard DHSV

Cement evaluation : green color refers to good cement isolation, pink to doubtful cement isolation sheet

References

1. ISO 16530-1:2017, Petroleum and natural gas industries -Well Integrity-Part 1: Life cycle governance Definitions Subsection 3.64
2. SPE-224652-MS Novel ADNOC Offshore Experience in Managing Wells Beyond Design Life